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System for tire inflation

Abstract

A system for tire inflation including a drive mechanism defining a rotational axis, including an eccentric mass that offsets a center of mass of the drive mechanism from the rotational axis along a radial vector; a pump arranged radially distal the rotational axis of the drive mechanism, including a chamber defining a chamber lumen, and a reciprocating element arranged at least partially within the chamber lumen and translatable along a pump axis; a drive coupler coupled between the drive mechanism at a first position and the reciprocating element at a second position fixed to the reciprocating element; a torque regulation mechanism; and a controller, communicatively coupled to the torque regulation mechanism; wherein the system is operable between at least a first mode and a second mode by the torque regulation mechanism in cooperation with the controller.

Inventors:	Richardson; Brandon (Burlingame, CA)
Applicant:	Aperia Technologies, Inc. (Hayward, CA)
Family ID:	74343260
Assignee:	Aperia Technologies, Inc. (Hayward, CA)
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Primary Examiner: Plakkoottam; Dominick L

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/097,479 filed 16 Jan. 2023, which is a continuation of U.S. patent application Ser. No. 17/868,311, filed 19 Jul. 2022, which is a continuation of U.S. patent

application Ser. No. 17/061,313, filed 1 Oct. 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/276,998 filed 15 Feb. 2019, which is a divisional of U.S. patent application Ser. No. 15/696,816 filed 6 Sep. 2017 which claims the benefit of U.S. Provisional Application Ser. No. 62/383,910, filed 6 Sep. 2016, and U.S. Provisional Application Ser. No. 62/519,061, filed 13 Jun. 2017, each of which is incorporated herein in its entirety by this reference. (2) U.S. patent application Ser. No. 17/061,313, filed 1 Oct. 2020, is also a continuation-in-part of U.S. patent application Ser. No. 16/161,771 filed 16 Oct. 2018, which is a continuation of U.S. application Ser. No. 15/280,737 filed 29 Sep. 2016, which claims the benefit of U.S. Provisional Application No. 62/235,121 filed 30 Sep. 2015 and is a continuation-in-part of U.S. application Ser. No. 14/839,009 filed 28 Aug. 2015, which is a continuation of U.S. application Ser. No. 14/198,967 filed 6 Mar. 2014, which is a continuation of U.S. application Ser. No. 14/019,941 filed 6 Sep. 2013, which is a continuation of U.S. application Ser. No. 13/797,826 filed 12 Mar. 2013, each of which are incorporated in their entireties by this reference.

TECHNICAL FIELD

(1) This invention relates generally to the pumping field, and more specifically to a new and useful tire-mounted pumping system in the pumping field.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a schematic representation of the tire inflation system.
- (2) FIG. 2 is a schematic representation of a variation of the tire inflation system.
- (3) FIGS. 3A-3C are schematic representations of example configurations of the tire inflation system.
- (4) FIGS. 4A-4B are top views of example relative configurations of the drive mechanism and the torque regulation mechanism.
- (5) FIG. 5 is a schematic of an example configuration of a slotted cam configuration of the drive coupler of the tire inflation system.
- (6) FIG. 6 is an exploded view of a variation of the tire inflation system.
- (7) FIGS. 7A and 7B are schematic representations of a variation of the stabilizing mechanism in the pumping and non-pumping modes, respectively.
- (8) FIGS. 8A and 8B are schematic representations of a second variation of the stabilizing mechanism in the pumping and non-pumping modes, respectively.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

(10) 1. Overview

(11) As shown in FIG. 1, the system for tire inflation **100** includes a drive mechanism **120**, a primary pump **130**, a drive coupler **140**, and a torque regulation mechanism **150**. The system can optionally include an energy storage device **160**, one or more sensors **170**, and a controller **180**. In a variation, the drive mechanism **120** includes a cam **122** and an eccentric mass **121**, the primary pump **130** includes a reciprocating element **131** and a pump body **132**, and the torque regulation mechanism **150** includes a first portion **151** (e.g., a stator) and a second portion **152** (e.g., a rotor).

(12) The system functions to inflate a tire. The system can also function to translate rotational motion into reciprocating linear motion that can be used to drive a tire inflator (e.g., a pump). The system can also function to translate relative motion between the primary pump **130** and cam **122** into a pumping force, wherein the eccentric mass **121** retains the cam **122** position relative to a gravity vector while the primary pump **130** rotates relative to the cam **122**. The system can be

operable between several modes, including a pumping (e.g., active) mode and a freewheeling (e.g., passive) mode. In the pumping mode, the tire inflation system **100** preferably pumps an external fluid, such as air, into the tire interior. The external fluid is preferably received from a first reservoir **910** (e.g., the external environment, a canister, etc.) during a recovery stroke of the primary pump **130** and pumped into a second reservoir (e.g., the tire) during a compression stroke of the primary pump **130**. However, the fluid can be otherwise suitably pumped. The first reservoir **910** is preferably the ambient atmosphere at a first pressure, and the second reservoir is preferably the tire interior (e.g., bladder) at a second pressure higher than the first pressure. However, the first and second reservoirs can be any other suitable fluid reservoirs at any other suitable absolute and/or relative pressures. In further alternatives, the fluid can be a fluid other than air (e.g., liquid water, pure nitrogen, etc.).

(13) In a first variation of the freewheeling mode, the eccentric mass **121** of the drive mechanism **120** rotates at substantially the same velocity as the wheel (and, thus, as the primary pump **130**) such that a negligible (e.g., zero, substantially zero) drive force is supplied by the drive mechanism **120** to the primary pump **130**. In the first variation, the torque regulation mechanism **150** can supply a torque to the eccentric mass **121** to excite the mass into rotation about a rotation axis (e.g., of the wheel hub) at the same velocity (e.g., substantially the same, exactly the same) as the wheel, whereupon angular momentum of the eccentric mass **121** substantially maintains the eccentric mass **121** in rotation. Upon excitation of the eccentric mass **121** into rotation at the same velocity as the wheel, the torque regulation mechanism **150** can cease supplying the torque. However, in an alternative implementation of the first variation of the freewheeling mode, the torque regulation mechanism **150** can supply a torque (e.g., continuously, periodically, asynchronously, etc.) to maintain the eccentric mass **121** in rotation about the rotation axis at a suitable velocity such that a negligible drive force is supplied by the drive mechanism **120** to the primary pump **130**.

(14) In a second variation of the freewheeling mode, the eccentric mass **121** can be statically connected to the system housing **110** and/or wheel. In the second variation of the freewheeling mode, the eccentric mass **121** rotates along with the housing **110** at the wheel speed, acted upon by a mechanical force supplied by the static connection. For example, the eccentric mass **121** can be clipped, latched, buckled, snapped, or otherwise suitably fastened to the housing **110** and/or any portion of the system or system-related component rotating along with the wheel (e.g., in a reference frame rotating at the same angular velocity as the wheel). However, the eccentric mass **121** can be otherwise suitably statically connected to the system housing **110** and/or wheel in the second variation of the freewheeling mode.

(15) In a third variation of the freewheeling mode, the eccentric mass **121** can be rotationally decoupled from (e.g., rotates freely relative to) the system housing **110** and/or wheel. In the third variation of the freewheeling mode, rotation of the eccentric mass **121** (e.g., when rotationally decoupled) does not supply a drive force to the primary pump **130** via the cam **122** and drive coupler **140**. For example, the system can include a clutch that can engage and disengage the eccentric mass **121** from the cam **122**, wherein the eccentric mass **121** and the cam **122** are disengaged during system operation in the freewheeling mode, and engaged in the pumping mode. In another example, the system can include a clutch that can engage and disengage the cam **122** from the drive coupler **140**, wherein the cam **122** and the drive coupler **140** are disengaged during system operation in the freewheeling mode, and engaged in the pumping mode. However, in the third variation of the freewheeling mode, the system can additionally or alternatively include any suitable mechanism for rotationally decoupling the drive mechanism **120** from the primary pump **130**.

(16) In a fourth variation of the freewheeling mode, the eccentric mass **121** is maintained at a hanging angle of substantially zero degrees relative to a gravity vector, such that no reciprocating action is produced by the cam **122** upon the reciprocating element **131** of the primary pump **130**. In the fourth variation, the eccentric mass **121** is preferably maintained at a zero hanging angle by the

torque regulation mechanism **150**, but can additionally or alternatively be otherwise suitably maintained at a zero hanging angle (e.g., by a locking mechanism).

(17) The system is preferably operable between the pumping and freewheeling modes by the torque regulation mechanism **150** in cooperation with the controller **180**. Controller **180** operation can include generating control instructions based on any suitable control algorithm, and incorporating any suitable sensor inputs. The control instructions can be generated in real-time, near-real time, or at any suitable time. The control instructions and/or parameter values thereof can be selected (e.g., from a database) based on the sensor input values or patterns (e.g., eccentric mass angular kinematics, system lateral kinematics, vehicle kinematics, etc.), calculated (e.g., target operation values calculated based on the sensor input values), optimized (e.g., for pumping, energy harvesting, eccentric mass rotational frequency, etc.), or otherwise determined. However, the system can be otherwise suitably operable between any suitable operating modes by any suitable control and/or regulation mechanism.

(18) The tire inflation system **100** preferably discontinuously inflates the tire (e.g. via periodic inflation, wheel speed-controlled inflation, actively controlled inflation, pressure-dependent inflation, etc.), but may continuously inflate the tire. The tire inflation system **100** is preferably powered by a direct mechanical linkage to the rotating wheel, such that the inflation system pumps fluid into the tire when the tire rotates; but the tire inflation system **100** can alternatively be powered by an actuator that is decoupled from the rotation of the tire (e.g., an electric motor having a separate power source). The tire inflation system **100** can pump fluid using a diaphragm system, a peristaltic system, a piston system, or any other suitable pumping mechanism. The tire inflation system **100** preferably mounts to a wheel (e.g. to the hub of a wheel), and preferably connects to the tire interior through a valve of the tire. The tire inflation system **100** is preferably configured to be mounted to the wheel for an extended period of time (e.g., on the order of weeks, months, or years); accordingly, removal of the tire inflation system **100** for routine tire pressure checks can be omitted.

(19) 2. Benefits

(20) Variants of the systems and/or methods can confer several benefits and/or advantages. First, variants of the system can provide improved resistance to entering an undesired spin condition in which the eccentric mass **121** rotates at substantially the same angular velocity as the wheel (e.g., the freewheeling mode) when reciprocal pumping is desired, by providing torque input to the eccentric mass **121** via the torque regulation mechanism **150**. The torque input can be modulated to provide a counter-force to torque ripple caused by the reciprocating pump (e.g., a back torque acting upon the eccentric mass **121**), to provide a counter-force to large back torques produced by the primary pump **130** during the compression stroke when the system is operated at low vehicle speeds and/or starting from a stopped state (e.g., zero velocity), and/or to provide a counter-force to transient forces resulting from road and/or driving irregularities (e.g., bumps, undulations, vehicle acceleration and deceleration, etc.). This can, in turn, actively increase the amount of time during driving in which the system can usefully pump air using energy harvested from the eccentric mass **121**.

(21) Second variants of the system can enable the tire inflation system **100** to be controllable (e.g., actively controllable) between the pumping and freewheeling modes, by transitioning the pendulum (e.g., eccentric mass **121**) into the freewheeling mode (e.g., wherein the pendulum is rotating at the wheel rotation speed) during periods in which the tire does not require inflation. By operating in the freewheeling mode during periods in which the tire(s) do not require inflation, wear on system components (e.g., reciprocating pump components) can be reduced and the maintenance-free system lifetime can be thus increased. The torque regulation mechanism **150** (e.g., in cooperation with a control system) can also actively transition the system into the pumping mode, by providing a torque against the eccentric mass **121** to control the hanging angle of the eccentric mass **121** relative to a gravity vector (e.g., to stop the eccentric mass **121** from rotating at the wheel rotation

speed). In one variation, this includes: determining the wheel rotation speed and controlling electric motor rotor rotation to substantially match the wheel rotation speed. In a second variation, this includes: determining the eccentric mass **121** angle relative to the gravity vector, determining a desired angle, and controlling the electric motor (e.g., the electric motor rotation speed, the angular position of the electric motor, the current or voltage supplied to the electric motor, etc.) to adjust the eccentric mass **121** angle to substantially match the desired angle. However, the system can be otherwise transitioned into the pumping mode. By transitioning into the pumping mode without relying on a passive exit from the freewheeling mode (e.g., due to normal perturbations to the rotary motion arising from road surface irregularities and/or driving behavior), fluid can be provided to the tires on demand (e.g., when tires require immediate inflation, imminent inflation, etc.).

(22) Third, variants of the system can confer several benefits related to on-demand, real-time tire inflation. Tires that are properly inflated improve vehicle fuel economy, and have longer lifetimes before replacement becomes necessary. Adjustable tire pressure in real or near-real time also enables adjustment of tire parameters (e.g., compressibility) to road and/or environmental conditions. For example, the tire pressure can be increased to take advantage of reduced rolling resistance on recently paved, smooth roads where the risk of a tire rupture due to road roughness is low. In another example, tire pressure can be automatically adjusted to account for changes in ambient pressure and/or temperature, such that an optimal pressure difference between the interior and exterior of the vehicle tire(s) is maintained.

(23) Fourth, variants of the system can be distributed at each wheel of the vehicle (or a subset of wheels of the vehicle), which can reduce the cost of an auto-inflation system compared to a centralized inflation system and can enable the control of tire pressure on a per-wheel basis without the need for complex and expensive plumbing, valve networks, and/or pressurized fluid manifolds. Performing pressurization at the wheel-end can also reduce the likelihood of pressure system failure due to a reduction in the number of pressurized system components, which can be vulnerable to shock and vibration damage when routed beneath a vehicle.

(24) Fifth, variants of the system can be physically rugged, robust, and/or otherwise resilient to the harsh environment in the vicinity of the wheel due to exposure to road debris and other hazards. The placement of variants of the system at the wheel hub area provides a well-suited area for physically shielding system components between the wheel hub and an outer surface of the system housing **110**.

(25) However, the system and/or method can confer any other suitable benefits and/or advantages.

(26) 3. System

(27) As shown in FIG. 2, the tire inflation system **100** can include: a housing **110**; a drive mechanism **120** that includes a cam **122** and an eccentric mass **121**; a primary pump **130** that includes a reciprocating element **131**, a pump body **132**, a return mechanism **133**, and one or more inlets **134**; a drive coupler **140** that connects the drive mechanism **120** and the primary pump **130**; a torque regulation mechanism **150** that includes a rotor and a stator; an energy storage device **160** that includes an energy dissipation mechanism **161**; one or more sensors **170**; and a controller **180**. Variants of the system or components thereof can be similar to the system and/or components described in U.S. application Ser. No. 15/280,737, filed 29 Sep. 2016, incorporated herein in its entirety by this reference.

(28) 3.1 Housing

(29) The housing **110** functions to couple system components to a rotating surface **900** (e.g., the hub of a wheel of a vehicle). The housing **110** can also function to mechanically protect (e.g., shield) system components from road debris and other objects that can transiently impact the wheel during vehicle operation. The housing **110** can also function as a mounting substrate for visual indicators of system performance (e.g., for an LED that reports the system status). The housing **110** is preferably removably coupled to a rotating surface **900**, such as by way of removable fasteners

(e.g., nuts and bolts, screws, brackets, etc.); additionally or alternatively, the housing **110** can be permanently coupled to the rotating surface **900** (e.g., via welding, rivets, permanent fasteners, etc.). The housing **110** is preferably coupled to a rotating surface **900** of the vehicle (e.g., that rotates during vehicle locomotion), and is more preferably coupled to the hub of a vehicle wheel. However, the housing **110** can additionally or alternatively be statically coupled to the rim of a vehicle wheel, a hubcap, to an axle of the vehicle, or any other suitable rotating or non-rotating surface **900** of the vehicle. The housing **110** is preferably coupled to the vehicle by way of a set of fasteners (e.g., arranged to mate with an existing bolt pattern of the wheel hub), but can additionally or alternatively be integrated directly into the vehicle (e.g., manufactured as part of the wheel hub or vehicle axle) or otherwise suitably attached to the vehicle by any other suitable mechanism. In a specific example, the housing **110** is contiguous with a hubcap of the wheel, and is fastened to the wheel (e.g., via a set of lugnuts) and functions both as a hubcap and the system housing **110**. The housing **110** is preferably rotatably coupled to the drive mechanism **120** (e.g., such that the eccentric mass **121** can rotate relative to the housing **110**) and statically coupled to the pump body **132** of the primary pump **130** (e.g., such that the primary pump **130** rotates with the housing **110** as the wheel rotates. Alternatively, the housing **110** can be statically coupled to the drive mechanism **120** and rotatably coupled to the primary pump **130**, or have any other suitable coupling to the other system components. The housing **110** is preferably substantially rigid, but can additionally or alternatively be flexible, resilient, or have any other suitable structural characteristics. The housing **110** is preferably substantially impermeable to fluids (e.g., waterproof) and can preferably at least partially shield system components from exposure to external liquids (e.g., water splashed onto the wheel from the road surface), but can additionally or alternatively be permeable.

(30) In a first specific example, as shown in FIG. **6**, the housing **110** includes an inner housing that includes a first portion **111** (e.g., mounting plate) that defines a hole pattern **1110** arranged to mate with an existing bolt pattern of the wheel, and a second portion **112** that mates with the first portion **111** to cooperatively define a housing lumen **113**. The housing lumen **113** contains the primary pump **130**, and defines an orifice **114** through which the pump can be connected to a first reservoir **910** of fluid (e.g., ambient air) and a second reservoir of fluid (e.g., the interior of a tire). The housing lumen **113** further contains the torque regulation mechanism **150**, which is disposed adjacent to a portion of the drive mechanism **120** such that a torque can be applied by the torque regulation mechanism **150** to the drive mechanism **120** and thereby adjust the angular position of the eccentric mass **121** of the drive mechanism **120** (e.g., to transition the system into the pumping mode or freewheeling mode). In this first specific example, the housing **110** further contains the cam **122**, and the eccentric mass **121** is arranged external to the inner housing and coupled to the cam **122** by an axle that extends through the second portion **112**. The eccentric mass **121** in this example extends radially past a furthest radial extent of the inner housing, and defines a portion along an arcuate section of the rotational path of the eccentric mass **121** that extends axially toward the first portion in of the inner housing. The housing **110** in this example can further include an outer housing that encloses the eccentric mass and the inner housing.

(31) In a second specific example, the housing **110** is integrated directly with a hubcap of a vehicle wheel, and defines a housing lumen **113**. The housing lumen **113** contains the primary pump **130**, the drive mechanism **120**, the torque regulation mechanism **150**, and the drive coupler **140**, and is substantially sealed against the external environment. The housing **110** defines an inlet, which can include a shielded cover (e.g., to prevent foreign matter besides air from entrance), through which the primary pump **130** draws ambient air for compression and pumping during system operation. In this second specific example, the eccentric mass **121** is arranged internal to the housing **110**. The eccentric mass **121** in this example extends radially toward an inner surface of the housing lumen **113**, and defines a portion along an arcuate section of the rotational path of the eccentric mass **121**.

(32) 3.2 Drive Mechanism

(33) The drive mechanism **120** of the tire inflation system **100** functions to generate a pumping force to drive the primary pump **130**. The drive mechanism **120** can also function to control the magnitude of the pumping force. The drive mechanism **120** preferably includes an eccentric mass **121** and a cam **122**, but can include any other suitable components for generating the pumping force (e.g., a rotary pump, a diaphragm pump, a turbopump, etc.). The pumping force generated by the drive mechanism **120** is preferably applied in a radial direction relative to the rotational axis **123** of the drive mechanism **120** (e.g., the rotational axis **123** of the wheel), but can alternatively be applied in any suitable direction. The pumping force is preferably applied cyclically (e.g., in a reciprocal manner to drive a reciprocating pump), but can additionally or alternatively be a constant force, a steadily increasing or decreasing force, or have any other suitable temporal profile.

(34) The drive mechanism **120** can be rotatably coupled to the housing **110**, such that the drive mechanism **120** is substantially stationary in a translating reference frame (e.g., translating with the vehicle) as the housing **110** and wheel rotate. The drive mechanism **120** preferably defines a rotational axis **123** about which a portion of the drive mechanism **120** can rotate, and more preferably the cam **122** of the drive mechanism **120** rotates about the rotational axis **123**. However, the rotational axis **123** can additionally or alternatively include the rotational axis **123** about which the eccentric mass **121** rotates, and/or any other suitable axis. The rotational axis **123** of the drive mechanism **120** is preferably coaxial with a rotational axis **123** of the tire inflation system **100** as a whole (e.g., the wheel rotational axis **123**), but can alternatively be offset (e.g., radially offset). The drive mechanism **120** preferably defines a single rotational axis **123** (e.g., about which the cam **122** and eccentric mass **121** rotate), but can alternatively define multiple rotational axes (e.g., a first rotational axis **123** about which the eccentric mass **121** rotates, and a second rotational axis **123** distinct from the first rotational axis **123** about which the cam **122** rotates).

(35) The cam **122** of the drive mechanism **120** functions to mechanically control the magnitude of the pumping force. The cam **122** can also function to convert a torque received from the drive mechanism **120** to a linear force, and apply the linear force against the reciprocating element **131** of the primary pump **130** during the compression stroke. The torque received and/or the linear force applied can be, in variations, constant in time, variable in time, adjustable, or have any other suitable characteristics. In a first variation, the torque provided is modulated in response to a back torque from the reciprocating pump (e.g., assisted by the torque regulation mechanism **150**, defined by a feature of the cam **122**, etc.). The cam **122** preferably defines a bearing surface **1220**, which can be an interior surface of the cam **122**, an exterior surface of the cam **122**, or any suitable combination of interior and exterior surfaces. The bearing surface **1220** can be continuous or discontinuous. In a specific example, as shown in FIG. 5, the bearing surface **1220** is defined within an interior of the cam **122** and includes a slotted lumen **1221**. However, the system can include any suitable cam **122** with any suitable configuration.

(36) The bearing surface **1220** can include a profile that, in variations, defines an arcuate surface, a surface having a non-uniform curvature, a uniform curvature, and/or any other suitable spatial profile. The profile of the bearing surface **1220** preferably controls the magnitude of the pumping force throughout the compression stroke (e.g., a modulated pumping force, a constant pumping force, etc.). The bearing surface **1220** is preferably arcuate, and preferably has a non-uniform curvature (e.g., an oblong profile or a reniform profile). Alternatively, the bearing surface **1220** can have a uniform curvature (e.g., a circular profile), an angular profile, or any other suitable profile. The bearing surface **1220** preferably includes a compression portion and a recovery portion, corresponding to the compression stroke and the recovery stroke of the primary pump **130**, respectively. The compression portion is preferably continuous with the recovery section, but can alternatively be discontinuous. The bearing surface **1220** preferably has a first section having a high curvature (preferably positive curvature or convex but alternatively negative curvature or concave) adjacent a second section having low curvature (e.g., substantially flat or having negative curvature compared to the first section). The bearing surface **1220** preferably additionally includes

a third section connecting the first and second sections, wherein the third section preferably provides a substantially smooth transition between the first and second sections by having a low curvature adjacent the first section and a high curvature adjacent the second section. The compression portion preferably begins at the end of the second section distal the first section, extends along the third section, and ends at the apex of the first section. The compression portion is preferably convex (e.g., when the bearing surface **1220** is an external bearing surface **1220**), but can alternatively be concave. The apex of the first section preferably corresponds to the top of the compression stroke (compressed position). The recovery portion preferably begins at the apex of the first section, extends along the second section, and ends at the end of the second section distal the first section. The recovery portion is preferably substantially flat or concave (e.g., when the bearing surface **1220** is an external bearing surface **1220**), but can alternatively be convex. The end of the second section preferably corresponds to the bottom of the recovery stroke (recovered position). The slope of the compression portion is preferably less than 30 degrees, but can alternatively have any suitable angle. When a roller is used as the force translator, the curvature of the bearing surface **1220** is preferably at least three times larger than the roller curvature or roller diameter, but can alternatively be larger or smaller. However, the bearing surface **1220** can have any suitable profile. The cam **122** is preferably substantially planar with the bearing surface **1220** defined along the side of the cam **122**, in a plane normal to the rotational axis **123** of the cam **122** (e.g., normal the broad face of the cam **122**). The bearing surface **1220** is preferably defined along the entirety of the cam **122** side, but can alternatively be defined along a portion of the cam **122** side. The generated pump force is preferably directed radially outward of the rotational axis **123**, more preferably along a plane normal to the rotational axis **123**. Alternatively, the cam **122** can have a rounded or otherwise profiled edge segment (transition between the cam **122** broad face and the cam **122** side), wherein the bearing surface **1220** can include the profiled edge. Alternatively, the arcuate surface is defined by a face of the cam **122** parallel to the rotational axis **123** of the cam **122**, wherein the generated pump force can be directed at any suitable angle relative to the rotational axis **123**, varying from parallel to the rotational axis **123** to normal to the rotational axis **123**. The compression portion preferably encompasses the majority of the cam **122** profile, but can alternatively encompass half the cam **122** profile or a small portion of the cam **122** profile. In one variation, the compression portion covers 315 degrees of the cam **122** profile, while the recovery portion covers 45 degrees of the cam **122** profile. However, the compression and recovery portions can cover any other suitable proportion of the cam **122** profile.

(37) The eccentric mass **121** (e.g., pendulum, offset mass) of the drive mechanism **120** functions to offset the center of mass of the drive mechanism **120** from the rotational axis **123** of the drive mechanism **120**. The offset functions to retain an angular position of the drive mechanism **120** relative to a gravity vector, in order to generate relative angular motion between the drive mechanism **120** and components statically coupled to the rotating surface **900** (e.g., the housing **110**, the pump body **132**, etc.). The eccentric mass **121** is preferably a homogenous (e.g., continuous) mass, but can additionally or alternatively be a heterogeneous (e.g., segmented, discontinuous, etc.) mass. In a specific example, as shown in FIG. 3B, the eccentric mass **121** is rotatably attached to the housing **110** at the rotation axis of the wheel and is distributed along a portion of an arc centered at the rotational axis **123**. The eccentric mass **121** is preferably a substantially singular, contiguous piece, but can alternatively be made up of multiple pieces and/or segments. In the latter case, the multiple pieces and/or segments are preferably substantially similar in shape, angular and radial position, and mass, but can alternatively be different in profile, mass, angular position, and/or radial position. The eccentric mass **121** can define a curved shape, flat surface, angular shape, and/or any other suitable geometry. At least a portion of the eccentric mass **121** preferably traces an arcuate section of the system perimeter (e.g., aligned with the hub perimeter, inset from the hub perimeter, outside the housing **110** perimeter, inside the housing **110** perimeter, etc.) such that a substantial fraction (e.g., between 10-90%, between 0-100%) of the

mass is distributed along the arcuate section. The arcuate section can include any suitable arc (e.g., 90°, 180°, etc.). However, in alternative variations, the eccentric mass **121** can be a spatially confined mass at an end of a pendulum that approximates a point mass. In some variants, the azimuthal distribution of the mass can be varied. For example, the eccentric mass **121** can include articulated arms that can be unfolded outward (e.g., automatically unfolded, manually unfolded, etc.) to distribute the mass along an arcuate section in the azimuthal direction about the rotational axis **123**. However, the eccentric mass **121** can be otherwise suitably configured and/or arranged. (38) The eccentric mass **121** is preferably curved, but can alternatively be substantially flat, angled, or have other suitable shape. The radius of the eccentric mass **121** curvature is preferably maximized, such that the eccentric mass **121** traces an arcuate section of the pump system perimeter. However, the eccentric mass **121** can have any other suitable curvature. The eccentric mass **121** preferably extends at least 90 degrees about the rotational axis **123** of the drive mechanism **120**, more preferably 180 degrees about the rotational axis **123**, but can extend more or less than 180 degrees about the rotational axis **123**. The eccentric mass **121** preferably has substantially more mass than the cam **122**, but can alternatively have a substantially similar mass or a smaller mass. The eccentric mass **121** preferably imparts 2 in-lb (0.225 Nm) of torque on the cam **122**, but can alternatively impart more or less torque.

(39) The eccentric mass **121** is preferably a separate piece from the cam **122**, and is preferably coupled to the cam **122** by a mass coupler **124**. Alternatively, the eccentric mass **121** can be incorporated into the cam **122**, wherein the eccentric mass **121** is incorporated along the perimeter of the cam **122**, incorporated into a half of the cam **122**, or incorporated along any other suitable portion of the cam **122**. The eccentric mass **121** can be statically coupled to the cam **122** or rotatably coupled to the cam **122**. In the variation wherein the eccentric mass **121** is statically coupled to the cam **122**, the eccentric mass **121** can be coupled to the cam **122** at the rotational axis **123** of the cam **122**, at the rotational axis **123** of the drive mechanism **120**, offset from the rotational axis **123** of the cam **122**, or at any other suitable portion of the cam **122**. The eccentric mass **121** can be permanently connected to the cam **122**. Alternatively, the eccentric mass **121** can be transiently connected (removably coupled) to the cam **122**, wherein the eccentric mass **121** can be operable between a pumping mode wherein the eccentric mass **121** is coupled to the cam **122** and a non-pumping mode wherein the eccentric mass **121** is disconnected from the cam **122**. The mass coupler **124** preferably has a high moment of inertia, but can alternatively have a low moment of inertia. The mass coupler **124** is preferably a disk, but can alternatively be a lever arm, plate, axle, or any other suitable connection. The mass coupler **124** preferably couples to the broad face of the cam **122**, but can alternatively couple to the edge of the cam **122**, along the exterior bearing surface **1220** of the cam **122**, to the interior bearing surface **1220** of the cam **122**, to an axle extending from the cam **122** (wherein the cam **122** can be statically fixed to or rotatably mounted to the axle), or to any other suitable portion of the cam **122**. The mass coupler **124** can couple to the cam **122** by friction, by a transient coupling mechanism (e.g., complimentary electric or permanent magnets located on the cam **122** and mass coupler **124**, a piston, a pin and groove mechanism, etc.), by bearings, or by any other suitable coupling means. When the mass coupler **124** couples to the cam **122** by a transient coupling mechanism, the mass coupler **124** is preferably operable between a coupled mode, wherein the mass coupler **124** connects the eccentric mass **121** to the cam **122**, and a decoupled mode, wherein the mass coupler **124** disconnects the eccentric mass **121** from the cam **122**. The mass coupler **124** can additionally function as a shutoff mechanism, wherein the mass coupler **124** is switched from the coupled mode to the decoupled mode in response to the detection of a shutoff event (e.g., the reservoir pressure reaching a threshold pressure). In one variation, the mass coupler **124** is a disk located within the lumen defined by an interior bearing surface **1220** of the cam **122**, wherein the disk can rotate relative to the interior bearing surface **1220** in the decoupled mode and is coupled to the interior bearing surface **1220** by a friction element in the coupled mode (e.g., the mass coupler **124** acts as a clutch).

In another variation, the mass coupler **124** is rotatably mounted on an axle extending from the cam **122** by bearings, wherein the mass coupler **124** can be statically coupled to the cam **122** by one or more sets of magnets or pistons extending from the adjacent broad faces of the cam **122** and mass coupler **124**.

(40) 3.3 Primary Pump

(41) The primary pump **130** of the tire inflation system **100** functions to pressurize fluid with the pumping force generated by the drive mechanism **120**. The primary pump **130** preferably includes a reciprocating element **131** and a pump body **132**, and can optionally include a return mechanism **133** and one or more inlets **134**. However, the primary pump **130** can include any other suitable components. In variations, the primary pump **130** can function to pressurize the fluid by receiving a reciprocating linear force at the reciprocating element **131**. The primary pump **130** is preferably statically mounted to the housing **110**, wherein the housing **110** is statically coupled to a rotating surface **900** of the vehicle (e.g., the hub of a wheel). However, the primary pump **130** can additionally or alternatively be statically coupled to a surface that rotates relative to the rotating surface **900** (e.g., that is stationary in an external translating reference frame), such that relative motion is generated between the reciprocating element **131** of the primary pump **130** and the rotating surface **900**. The primary pump **130** is preferably positioned radially distal the rotational axis of the drive mechanism **120**, but can additionally or alternatively be positioned at least partially coaxially with the rotational axis of the drive mechanism **120** or otherwise suitably arranged. The position of the primary pump **130** relative to the drive mechanism **120** can be fixed or adjustable (e.g., manually adjustable, automatically adjustable, etc.).

(42) In a first variation, the primary pump **130** includes a positive displacement pump wherein the reciprocating element **131** is a piston, and defines a pump cavity (e.g., pump lumen, cylinder) within the pump body **132**. In a specific example of this variation, the primary pump **130** is a reciprocating piston pump. In a second variation, the primary pump **130** includes a peristaltic pump. However, the primary pump **130** can include any other suitable pumping mechanism.

(43) The reciprocating element **131** of the primary pump **130** functions to translate back and forth in a reciprocating manner within the pump body **132** to compress fluid transferred from the first reservoir **910** to the second reservoir (e.g., to the tire). The reciprocating element **131** can also function to receive the pumping force from the cam **122** and translate within the lumen of the pump, actuating relative to the pump body **132**. This actuation preferably creates a variable pressure within the lumen. The reciprocating element **131** is preferably operable between a compressed position and a recovered position. In the compressed position, a portion of the reciprocating element **131** (e.g., the center) is preferably proximal the pump body **132** bottom. In the recovered position, the portion of the reciprocating element **131** is preferably distal the pump body **132** bottom, and is preferably proximal the pump body **132** opening. The reciprocating element **131** preferably travels along a compression stroke to transition from the recovered position to the compressed position, and travels along a recovery stroke to transition from the compressed position to the recovered position. The reciprocating element **131** can additionally be positioned at a pressurized position, wherein the reciprocating element **131** is located at a second position distal the pump body **132** bottom, wherein the second position is further from the pump body **132** bottom than the recovered position. The reciprocating element **131** is preferably at the pressurized position when the force provided by the lumen pressure exceeds the force provided by the cam **122** on the reciprocating element **131**.

(44) The reciprocating element **131** preferably translates along an actuation axis within the primary pump **130** throughout the compression stroke, and can additionally translate along the actuation axis throughout the recovery stroke. The reciprocating element **131** preferably includes an actuating area that provides the pressurization force. The actuating area is preferably the surface area of a broad face of the reciprocating element **131**, more preferably the surface area of the broad face proximal the lumen but alternatively any other suitable broad face. Alternatively, the actuating

area can be the surface area of a section of the reciprocating element **131** that translates between the compressed position and the recovered position (e.g., the center portion).

(45) The reciprocating element **131** preferably forms a fluid impermeable seal with the pump body **132**, more preferably with the walls defining the pump body **132** opening, such that the reciprocating element **131** substantially seals the pump body **132** opening. The reciprocating element **131** can be sealed to the pump body **132** by a retention mechanism. The retention mechanism is preferably a clamp that applies a compressive force against the reciprocating element **131** edge and the pump body **132** wall, but can alternatively be screws or bolts through the reciprocating element **131** edge, adhesive between the reciprocating element **131** and the pump body **132** wall or over the reciprocating element **131** and the pump body **132** wall, or any other suitable retention mechanism. The reciprocating element **131** can also be sealed against the pump body **132** wall by melting the interface between the reciprocating element **131** and pump body **132** wall, or by any other suitable means of sealing the reciprocating element **131** against the pump body **132** wall.

(46) The reciprocating element **131** is preferably a flexible diaphragm, but can alternatively be a substantially rigid piston, a piston coupled to the diaphragm, or any other suitable element that actuates in response to the pumping force. The diaphragm is preferably a rolling diaphragm (e.g., with a rolled perimeter, wherein the diaphragm is preferably coupled to the pump body **132** with the extra material distal the lumen) but can also be a flat diaphragm, a domed diaphragm (preferably coupled to the pump body **132** with the apex distal the lumen, but alternatively coupled to the pump body **132** with the apex proximal the lumen), or any other suitable diaphragm.

(47) The pump body **132** functions to cooperatively compress fluid along with the reciprocating element **131**. The pump body **132** defines a lumen (e.g., cylinder cavity) in which the fluid is compressed. The pump body **132** is preferably statically mounted to the housing **110**, but can be otherwise suitably arranged relative to the housing **110** and/or other system components.

(48) The primary pump **130** can include a return mechanism **133**, which functions to bias the reciprocating element **131** in the reverse direction to the direction of the compression stroke during the recovery stroke. The return mechanism **133** preferably provides a recovery force that is less than the compression force provided by the third section of the cam **122**, but larger than the force applied by the cam **122** in the second section. The recovery force is preferably provided in a direction substantially parallel to a radial vector extending from the rotational axis of the drive mechanism **120**, but can alternatively be provided in any suitable direction. The return mechanism **133** is preferably located on the pump body **132** side of the reciprocating element **131** (distal the cam **122** across the reciprocating element **131**), wherein the return mechanism **133** preferably pushes the reciprocating element **131** from the compressed position, through the recovery stroke, and to the recovered position. Alternatively, the return mechanism **133** can be located on the cam **122** side of the reciprocating element **131** (distal the pump body **132** across the reciprocating element **131**), wherein the return mechanism **133** pulls the reciprocating element **131** back to the recovered position from the compressed position. The return mechanism **133** is preferably coupled to the perimeter of the reciprocating element **131** or to a component (e.g., a brace) coupled to the reciprocating element **131** and extending past the pump body **132** walls, but can alternatively be coupled to the body of the reciprocating element **131** (e.g., to the section actuating between the compressed position **222** and the recovered position). The return mechanism **133** is preferably coupled to the reciprocating element **131** external the pump body **132**, but can alternatively be coupled to the reciprocating element **131** within the pump body **132** **240**. The return mechanism **133** is preferably a spring, but can also include the intrinsic properties of the actuation element (e.g., the elasticity of the diaphragm) or any other suitable return mechanism **133**.

(49) The return mechanism **133** can, in further variations, include an internal spring, an exterior spring (e.g., mounted to an outer surface of the pump body **132**), a secondary cam **122** that drives the reciprocating element **131** in opposition to the cam **122** of the drive mechanism **120**, and/or any

other suitable mechanism.

(50) The primary pump **130** can include one or more inlets **134**, which function to receive fluid from the first reservoir **910** into the lumen of the pump body **132** for compression. The inlets **134** can be perpetually open (e.g., fixed orifice **114s** in the pump body **132**), actuatable (e.g., via controllable valves), shielded (e.g., to protect against influx of foreign matter besides the working fluid), or otherwise suitably constituted.

(51) 3.4 Drive Coupler

(52) The drive coupler **140** of the tire inflation system **100** functions to actuate the reciprocating element **131** of the primary pump **130** through the compression stroke as the primary pump **130** rotates about the rotational axis of the wheel. The drive coupler **140** can also function to translate the reciprocating element **131** through the recovery stroke. The drive coupler **140** is preferably coupled between the cam **122** of the drive mechanism **120** and the reciprocating element **131** of the primary pump **130**, but can alternatively be otherwise suitably coupled. In a first variation, the drive coupler **140** is coupled to the cam **122** by way of a roller bearing **141** captive within an oblong slot defined by the cam **122**, and pinned to the reciprocating element **131** (e.g., rotatable about a fixed point). In a second variation, the drive coupler **140** is pinned to both the cam **122** and the reciprocating element **131**. The drive coupler **140** preferably defines an axis having an arcuate position that is fixed relative to the arcuate position the primary pump **130** (e.g., the angular position of the drive coupler **140** about the rotational axis of the wheel is fixed relative to the angular position of the primary pump **130**). Preferably, the drive coupler **140** rotates with the primary pump **130** as both components rotate about the rotational axis of the wheel. However, the drive coupler **140** can additionally or alternatively exhibit a different relative rotation to the primary pump **130** (e.g., a different angular velocity, a different trajectory, an off-axis trajectory, etc.).

(53) 3.5 Torque Regulation Mechanism

(54) The torque regulation mechanism **150** functions to regulate the torque supplied to the drive mechanism **120** in order to transition the tire inflation system **100** between the pumping and freewheeling operation modes. The torque regulation mechanism **150** can also function to receive torque from the drive mechanism **120** and convert the received torque into electrical potential energy (e.g., to operate as a dynamo). The torque regulation mechanism **150** can also function to provide a torque (e.g., based on instructions from the controller **180**) to transition the tire inflation system **100** between the pumping mode and the freewheeling mode, and/or to maintain the tire inflation system **100** in one or more of the pumping mode, the freewheeling mode, and any other suitable operating modes. The torque regulation mechanism is preferably configured to apply a torque based on instructions received from a controller. The instructions can be automatically generated by the controller, generated by a system user in communication with the controller (e.g., manually via an electromechanical interface, wirelessly via a wireless transceiver, etc.), or otherwise suitably generated.

(55) The torque regulation mechanism **150** preferably includes a first portion **151** and second portion **152** that rotate relative to one another, but can be otherwise configured. In one variation, the first portion **151** includes a stator that is statically coupled to a rotating surface **900** (e.g., the housing **110** statically coupled to the wheel) and the second portion **152** includes a rotor that is statically coupled to the eccentric mass **121** such that the rotor rotates along with the eccentric mass **121**. In another variation, the stator is statically coupled to the eccentric mass **121** and the rotor is coupled to the rotating surface **900** by way of the housing **110**. The rotor and stator are preferably concentrically arranged, but can alternatively be offset (e.g., and mechanically linked by a force transfer mechanism **153**). However, the first and second portion **152s** of the torque regulation mechanism **150s** can be otherwise suitably relatively arranged. In a specific example, the torque regulation mechanism **150** is coupled to the eccentric mass **121** via an intermediate force transfer mechanism **153** (e.g., a gear, a gearbox, a belt, a chain, a clutch, etc.). The torque regulation

mechanism **150** is preferably electrically coupled to the controller **180** (e.g., to receive control instructions and/or signals) and the energy storage device **160** by way of one or more direct electrical power and/or data connections. However, the torque regulation mechanism **150** can be otherwise suitably coupled to the controller **180** and/or energy storage device **160**.

(56) The torque regulation mechanism **150** is preferably arranged at a different plane from the rotation plane of the eccentric mass **121** (e.g., distal the rotation plane of the eccentric mass **121** in a direction away from the wheel hub, distal the rotation plane of the eccentric mass **121** in a direction toward the wheel hub, etc.). As shown in FIG. 4A, the torque regulation mechanism **150** can be arranged toward the vehicle (e.g., toward the vehicle centerline) relative to the drive mechanism **120** (e.g., the eccentric mass **121** of the drive mechanism **120**). As shown in FIG. 4B, the torque regulation mechanism **150** can be arranged away from the vehicle relative to the drive mechanism **120**. However, the torque regulation mechanism **150** can additionally or alternatively be arranged in the same plane (e.g., coaxially arranged, offset from the rotation axis of the eccentric mass **121**, etc.). In a first variation, as shown in FIG. 3A, the torque regulation mechanism **150** is arranged coaxially with the rotation axis of the wheel and the eccentric mass **121**. In further variations, as shown in FIGS. 3B and 3C, the torque regulation mechanism **150** is arranged at an offset position from the rotation axis of the eccentric mass **121**, and connected to the eccentric mass **121** via a force transfer mechanism **153** (e.g., a chain and sprocket, a drive belt, etc.). However, the torque regulation mechanism **150** can be otherwise arranged relative to the drive mechanism **120**, axis of rotation, or eccentric mass **121**. The torque regulation mechanism **153** can apply a: radially inward force, radially outward force, linearly outward force (e.g., away from the wheel or longitudinal vehicle axis), linearly inward force (e.g., toward the vehicle), arcuate force (e.g., within the same plane as eccentric mass rotation), or any other suitable force to the eccentric mass, cam, pump, or other pumping component. The torque regulation mechanism can be statically mounted to: the housing (e.g., interior, exterior, component proximal the tire, component distal the tire, an arcuate segment of the sidewall, etc.), the eccentric mass, the cam, the pump, or to any suitable system component.

(57) The torque regulation mechanism **150** preferably includes an electric motor, but can additionally or alternatively include any suitable torque generation and/or regulation mechanism. The electric motor can be an outrunner motor, an inrunner motor, a brushed motor, a brushless motor, an alternating-current motor, a directcurrent motor, a permanent magnet motor, an induction motor, a servo motor, a stepper motor, and/or any other suitable motor. The electric motor preferably generates a rotational force, but can alternatively generate a linear force (e.g., be a linear actuator) or generate any suitable force. In variations, the torque regulation mechanism **150** can include mechanical torque regulation components, such as gears, springs, levers, and any other suitable clockwork components that do not require electrical energy for operation.

(58) The rotor of the torque regulation mechanism **150** functions to move relative to the stator under an applied electromotive force to generate a torque on components statically coupled to the rotor. The rotor can also function to move relative to the stator under an applied torque to generate an electromotive force that can be harvested and stored as electrical potential energy (e.g., at the energy storage device **160**). In a first variation, the rotor is statically coupled to a surface that rotates with the wheel. In a second variation, the rotor is statically coupled to a surface that is substantially stationary relative to the wheel. However, the rotor can be otherwise suitably coupled.

(59) The stator of the torque regulation mechanism **150** functions to move relative to the rotor under an applied electromotive force to generate a torque on components statically coupled to the stator. The stator can also function to move relative to the rotor under an applied torque to generate an electromotive force that can be harvested and stored as electrical potential energy (e.g., at the energy storage device **160**). In a first variation, the stator is statically coupled to a surface that is substantially stationary relative to the wheel. In a second variation, the stator is statically coupled to a surface that rotates with the wheel. However, the stator can be otherwise suitably coupled.

(60) The torque regulation mechanism **150** can include an engagement mechanism **154** that functions to mechanically engage and/or disengage the eccentric mass **121** from other system components. For example, the engagement mechanism **154** can include a clutch that mechanically engages the eccentric mass **121** and the drive coupler **140** during system operation in the pumping mode (e.g., such that a drive force is provided by the eccentric mass **121** when the eccentric mass **121** is maintained at a non-zero hanging angle), and that mechanically disengages the eccentric mass **121** and the drive coupler **140** during system operation in the freewheeling mode (e.g., such that no drive force is provided by the eccentric mass **121** irrespective of the angular position and/or velocity of the eccentric mass **121**). In some variations, the mass coupler **124** can function as an engagement mechanism **154**. However, the engagement mechanism **154** can include any other suitable mechanism for mechanically retaining the eccentric mass **121** relative to the pump and/or other rotating components of the system.

(61) In a first specific example, as shown in FIG. **6**, the torque regulation mechanism **150** includes an electric motor wherein the stator of the electric motor is rigidly attached to the eccentric mass **121** (e.g., an arcuate segment of the stator defines a portion of the eccentric mass **121**), the rotor of the electric motor is rigidly coupled to a rotating surface **900** (e.g., the housing **110**, the wheel hub, via mounting components, directly coupled via a weld, etc.), and the rotor is connected to the drive coupler **140** that drives the primary pump **130**. In a second specific example, the torque regulation mechanism **150** includes an electric motor wherein the stator is rigidly mounted to the housing **110**, and is offset from the tire inflation system **100**'s rotational axis and is connected to the eccentric mass **121** by a force linkage (e.g., a gearbox).

(62) 3.6 Energy Storage Device

(63) The tire inflation system **100** can include an energy storage device **160**, which functions to provide power to the torque regulation mechanism **150**. The energy storage device **160** can also function to receive power from the torque regulation mechanism **150** (e.g., when the torque regulation mechanism **150** is operating as a dynamo). The energy storage device **160** can, in some variations, function to store compressed fluid generated by the primary pump **130** (e.g., in a compressed air canister). The energy storage device **160** is preferably coupled to the torque regulation mechanism **150** (e.g., via a direct electrical connection for power provision and/or reception), but can additionally or alternatively be coupled to the controller **180**, primary pump **130**, and/or any other system components. The system preferably includes a single energy storage device **160**, but can additionally or alternatively include redundant (e.g., multiple) energy storage device **160**s (e.g., to provide backup power to system components such as the torque regulation mechanism **150**). The energy storage device **160** is preferably coupled to the housing **110** and rotates with the wheel, but can alternatively be coupled to the eccentric mass **121** or to any other suitable system component. The energy storage device **160** is preferably arranged axially inward (e.g., along the direction of the vehicle axle) from the eccentric mass **121**, but can alternatively be arranged axially outward from the eccentric mass **121**. In a first variation, the energy storage device **160** includes a battery. In further variations, the energy storage device **160** can include a super capacitor, a compressed air canister, one or more springs, and/or any other suitable energy storage mechanisms.

(64) The energy storage device **160** can optionally include an energy dissipation mechanism **161** that functions to dissipate excess energy generated by the torque regulation mechanism **150** (e.g., when the torque regulation mechanism **150** is operating as a dynamo) in cases wherein the energy storage device **160** is at full capacity (e.g., when the battery is fully charged). For example, the energy dissipation mechanism **161** can include an electrical resistor, a resistor network, and/or any other suitable passive component for dissipating electrical energy in variations wherein the energy storage device **160** includes an electrical energy storage device **160** (e.g., a battery, capacitor, supercapacitor, etc.). In another example, the energy dissipation mechanism **161** can include an active energy dissipation mechanism **161**, such as a fan, water pump, light emitting element, and/or

any other suitable powered mechanism, to utilize excess recovered energy stored at the energy storage device **160** (e.g., for the purpose of cooling, user notification generation, etc.).

(65) 3.7 Sensors

(66) The tire inflation system **100** can include one or more sensors **170**, which function to sense operational parameters of the system (e.g., tire pressure, whether the system is in an “on” state or an “off” state, whether the system is operating within nominal ranges, etc.). The sensors **170** can also function to provide sensor data to a controller **180**. The sensors **170** can also function to detect, in cooperation with the controller **180**, whether the system is operating in the freewheeling and/or pumping modes (e.g., by comparing a measured rotational velocity of the eccentric mass **121** with a measured rotational velocity of the wheel). The system can include one or more pressure sensors **170**, which can be connected to the output of the primary pump **130** to monitor the pressure of the fluid provided to the tire. The sensor(s) are preferably connected to the controller **180** (e.g., via a signal pathway) to provide sensor data (e.g., sensor signals) to the controller **180**, and mounted on and/or within the housing **110** (e.g., for mechanical support). However, the sensor(s) can be otherwise suitably connected. System sensors **170** can include pressure sensors **170** (e.g., capacitively-based diaphragm deflection gauges), flow rate sensors **170**, mass flow sensors **170** (e.g., inline impellers), orientation sensors **170** (e.g., accelerometers, inertial measurement units, gyroscopes, etc.), rotary encoders, and/or any other suitable type of sensor.

(67) In a first variation, the system includes a pressure sensor arranged at an interface between the output of the primary pump **130** and the inlet of the second reservoir (e.g., the tire) to continuously measure the static pressure of the second reservoir. In a second specific example, the system includes a rotary encoder coupled to the torque regulation mechanism **150** that periodically measures (e.g., at 1 kHz) the angular position of the rotor of the torque regulation mechanism **150**.

(68) In a second variation, the system includes a sensor (e.g., an angular position sensor, angular velocity sensor, etc.) communicatively coupled to the controller **180**, that detects and outputs the relative angular velocity between the eccentric mass **121** and the primary pump **130**, wherein the controller **180** generates instructions based on an output of the differential angular velocity sensor and controls the torque regulation mechanism **150** based on the instructions. The sensor can be arranged to be in contact with the eccentric mass **121** (e.g., a contact sensor) and output the angular position and/or velocity based a characteristic of the contact; for example, the electrical resistance through a portion of the sensor can increase or decrease based on the position at which the sensor contacts the eccentric mass **121**. The sensor can additionally or alternatively be physically separated from the eccentric mass **121**; for example, the sensor can include an optical sensor that counts the frequency of optical occlusions of the optical sensor by the eccentric mass **121** during rotation of the eccentric mass **121** and/or the wheel, from which the angular velocity of the eccentric mass **121** can be computed (e.g., by the controller **180**). However, the system can include any other suitable sensors **170**.

(69) 3.8 Controller

(70) The tire inflation system **100** can include a controller **180**, which functions to generate control inputs in response to received sensor data and/or instructions. The controller **180** can also function to control the torque regulation mechanism **150** to operate the system between operating modes (e.g., the pumping mode, the freewheeling mode, etc.). The controller **180** can also function to adjust the pressure set point of the tire inflation system **100** and to control the tire inflation system **100** to maintain the tire pressure at the pressure set point. The controller **180** can also function to generate messages in response to system behavior (e.g., error codes). The controller **180** can be communicatively coupled to the sensors **170** and torque regulation mechanism **150** of the system, and in some variations can be communicatively coupled to a remote computing system (e.g., a vehicle ECU, a mobile device within the vehicle, etc.) via a communication system (e.g., wired communications system; wireless communications system, such as Bluetooth, WiFi, Zigbee, cellular, etc.).

(71) In a first specific example of controller **180** operation, the controller **180** detects a perturbative torque to the eccentric mass **121** (e.g., from an angular position sensor of the eccentric mass **121**) and controls the torque regulation mechanism **150** (e.g., an electric motor) to dampen the perturbative torque and maintain the system in the pumping mode (e.g., maintain a non-zero angle between a gravity vector and the eccentric mass **121**) and prevent the perturbative torque from transitioning the system from the pumping mode into the freewheeling mode. The perturbative torque can be detected via heuristic comparisons (e.g., pattern matching), deterministic comparisons (e.g., an oscillation exceeding a threshold perturbation magnitude), and/or in any other suitable manner. The perturbative torque can be a back torque generated by the primary pump **130** (e.g., due to reciprocating pump dynamics), a disturbance originating from road roughness, external forces and/or shocks, or any other source of torque or force.

(72) In a second specific example of controller **180** operation, the controller **180** receives an instruction (e.g., from a vehicle control system, a user, etc.) to cease pumping (e.g., to transition the system into the freewheeling operating mode from the pumping mode), and in response controls the torque regulation mechanism **150** to apply a torque to the eccentric mass **121** to induce rotation of the eccentric mass **121** at substantially the same angular velocity of the rotating wheel.

(73) 3.9 System Examples

(74) In a first specific example of the tire inflation system **100**, the system includes a drive mechanism **120**, a pump, a drive coupler **140**, a torque regulation mechanism **150**, and a controller **180**. The drive mechanism **120** defines a rotational axis, and includes a cam **122** rotatable about the rotational axis and an eccentric mass **121** coupled thereto that offsets a center of mass of the drive mechanism **120** from the rotational axis along the radial vector. The pump is arranged radially distal the rotational axis of the drive mechanism **120**, and includes a chamber defining a chamber lumen, and a reciprocating element **131** arranged at least partially within the chamber lumen and translateable along a pump axis normal to the rotational axis. The drive coupler **140** is coupled between the cam **122** at a first position and the reciprocating element **131** at a second position. The first position is radially distal the rotational axis about which the cam **122** rotates, and the second position is fixed (e.g., pinned) to the reciprocating element **131**. The torque regulation mechanism **150** (e.g., an electric motor, a clockwork mechanism, etc.) includes a first portion rigidly coupled to the eccentric mass **121**, and a second portion rotatably coupled to the first portion. The controller **180** is communicatively coupled to the torque regulation mechanism **150** (e.g., by a hardwire data connection, serial data port, etc.), and to a sensor (e.g., an angular velocity sensor, an angular position sensor, a rotary encoder, etc.) that senses an angular characteristic (e.g., angular position, angular velocity, etc.) of the eccentric mass **121** relative to a gravity vector. The controller **180** is configured to operate the system between the pumping mode and the freewheeling mode. In the pumping mode, the torque regulation mechanism **150** maintains the eccentric mass **121** at a hanging angle (e.g., defined by the radial vector between the rotational axis and an end of the eccentric mass **121** opposing the point at which the eccentric mass **121** is connected to the rotational axis) greater than 0° relative to the gravity vector. In cases where the system is in the freewheeling mode prior to transitioning to the pumping mode, the torque regulation mechanism **150** applies a torque to the eccentric mass **121** to stimulate the eccentric mass **121** to exit a spin condition (e.g., wherein the eccentric mass **121** is rotating about the rotational axis at substantially the same angular velocity as the wheel to which the system is attached).

(75) In a related specific example, the system further includes an energy storage device **160** (e.g., a battery, a torsional spring, a pneumatic cylinder, etc.) communicatively coupled (e.g., via a direct electrical connection, a direct mechanical connection, a fluid connection, etc.) to the torque regulation mechanism **150** and the controller **180**. The energy storage device **160** is operably between a harvesting mode and a powering mode. In the harvesting mode, the energy storage device **160** receives and stores energy harvested from the torque regulation mechanism **150**, which in turn receives a torque input from the eccentric mass **121**. For example, in a case where the

eccentric mass **121** is decoupled from the cam **122** and/or primary pump **130** (e.g., such that zero drive force is provided to the reciprocating element **131**), the eccentric mass **121** can be maintained at a non-zero hanging angle relative to a gravity vector in order to harvest energy (e.g., gravitational energy) to store within the energy storage device **160**. In the powering mode, the energy storage device **160** provides energy to the torque regulation mechanism **150**, which in turn provides a torque input to the eccentric mass **121**. The system and the energy storage device **160** of this example is preferably operated between the harvesting and powering mode by way of the controller **180**, but can be additionally or alternatively operated by any suitable mechanism or control instructions.

(76) The systems and methods of the preferred embodiment and variations thereof can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions are preferably executed by computer-executable components preferably integrated with the system and one or more portions of the processor and/or the controller **180 430**. The computer-readable medium can be stored on any suitable computer-readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, or any suitable device. The computer-executable component is preferably a general or application specific processor, but any suitable dedicated hardware or hardware/firmware combination device can alternatively or additionally execute the instructions.

(77) In a second specific example, the tire inflation system **100** includes a drive mechanism **120**, a drive coupler **140**, a reciprocating pump, a torque regulation mechanism **150**, a housing **110**, and a controller **180**. The drive mechanism **120** defines a rotational axis, and includes a cam **122** rotatable about the rotational axis and an eccentric mass **121** coupled to the cam **122** that offsets a center of mass of the drive mechanism **120** from the rotational axis along a radial vector and is rotatable about the rotational axis. The cam **122** includes a slotted lumen that defines an interior surface. The drive coupler **140** defines a first and second end, and the first end of the drive coupler **140** is coupled to the cam **122** at a position radially distal the rotational axis. The first end further includes a roller bearing, and at the coupling position to the cam **122** the roller bearing is captivated within the slotted lumen and contacts the cam **122** at the interior surface. The second end is coupled to the reciprocating pump. The reciprocating pump is arranged radially distal the rotational axis of the drive mechanism **120**, and includes a pump body **132** (e.g., chamber) that defines a chamber lumen, and a reciprocating element **131** arranged at least partially within the chamber lumen and translatable along a pump axis defined longitudinally along the chamber lumen and is normal to the rotational axis (i.e., the pump axis is perpendicular to the rotational axis of the system). The reciprocating pump further includes a return mechanism **133** (e.g., a spring) that applies a return force (e.g., a spring force) that biases the reciprocating element **131** along the pump axis towards the uncompressed position (e.g., away from the base of the chamber lumen). The return mechanism **133** is fixed to the reciprocating element **131** and the chamber, and is arranged externally to the chamber lumen (e.g., on the outside of the pump body **132**). The torque regulation mechanism **150** applies a controllable torque to the eccentric mass **121**, such that the controllable torque urges rotation of the eccentric mass **121** about the rotational axis. The torque regulation mechanism **150** further includes an engagement mechanism **154** (e.g., a clutch) that engages the eccentric mass **121** in at least a first and second configuration. In the first configuration, the eccentric mass **121** is mechanically coupled to the primary pump **130** (e.g., via the cam **122** and the drive coupler **140**), whereas in the second configuration, the eccentric mass **121** is mechanically decoupled from the primary pump **130** (e.g., via rotational decoupling from the cam **122**, mechanical decoupling from the drive coupler **140**, etc.). The torque regulation mechanism **150** further includes an electric motor and a force transfer mechanism **153** (e.g., a gearbox), and the force transfer mechanism **153** is coupled between the electric motor and the eccentric mass **121** such that torques transmitted between the electric motor and the eccentric mass **121** are mechanically transferred through the

gears of the gearbox. The electric motor is arranged at an offset position from the rotational axis, and the rotational axis of the electric motor (e.g., defined by an output shaft of the motor) is parallel to the rotational axis of the eccentric mass **121**. The housing **110** retains the pump, the torque regulation mechanism **150**, and the cam **122**. The eccentric mass **121** is arranged external to the housing **110** and coupled to the cam **122** by a fixed axle that extends through an orifice **114** of the housing **110** along the rotational axis. A portion of the eccentric mass **121** is rotatable about the rotational axis along a circular path, the circular path having a radius greater than the farthest radial extent of the housing **110** (e.g., outside the perimeter of the housing **110**). The eccentric mass **121** is distributed along an arcuate section of the circular path (e.g., a 90° section, a 180° section, etc.).

(78) In the second specific example above, the controller **180** is communicatively coupled to the torque regulation mechanism **150** (e.g., via a direct electrical connection, a serial data connection, a parallel data connection, a wireless data connection, etc.) and generates instructions which are provided (e.g., by the controller **180**) to the torque regulation mechanism **150** to operate the system between a first and second mode. In the first mode, the drive coupler **140** is mechanically engaged with the eccentric mass **121**, such that a drive force is provided to the primary pump **130** by the relative motion between the eccentric mass **121** and the primary pump **130**. In the second mode, the drive coupler **140** is mechanically disengaged from the eccentric mass **121**, such that no drive force is provided to the primary pump **130** by the relative motion (e.g., in cases where relative motion occurs) between the eccentric mass **121** and the primary pump **130** or by any other kinematic mechanism.

(79) Although omitted for conciseness, the preferred embodiments include every combination and permutation of the various system components and/or method blocks.

(80) The system **100** can additionally include a stabilizing mechanism **700** that functions to reduce rotational surface imbalance when the eccentric mass **121** becomes excited (e.g., begins spinning) when the system **100** rotates at or near the excitation frequency of the eccentric mass **121**. The stabilizing mechanism **700** is preferably the eccentric mass **121**, wherein the eccentric mass **121** is collectively formed from multiple sections. However, the stabilizing mechanism **700** can alternatively be any other suitable stabilizing mechanism **700**. When the eccentric mass **121** begins to spin, the composite sections of the eccentric mass separate. This is particularly useful when system oscillations cause the eccentric mass **121** (and positioning mechanism) to spin about the shaft; the centrifugal forces cause the sections of the split eccentric mass **121** to separate and be evenly distributed about the axis of system rotation, as shown in FIGS. 7A and 7B. Not only does this have the effect of dynamically balancing the system and/or rotating surface **900**, but the even distribution of the eccentric mass **121** within the system also halts system pumping. The latter effect can allow the eccentric mass **121** to additionally function as a control mechanism, wherein the eccentric mass resonant frequency can be tailored such that pumping is ceased when a predetermined rotation speed or vibration frequency is reached. The multiple sections are preferably each positioned the same radial distance away from the rotational axis (the eccentric mass **121** is radially divided into multiple sections, wherein the multiple sections have different angular positions), but can alternatively be positioned at different radial distances (e.g., wherein the multiple sections have substantially similar angular positions, etc.). The multiple sections preferably share a common plane, wherein the common plane is preferably substantially parallel to the rotational surface. The multiple sections can collectively form an arc, centered about the rotational axis, that intersects the common plane (e.g., the multiple sections are adjacent along an arc), form a block that intersects the common plane, or collectively form any other suitable structure. Alternatively, the multiple sections can be stacked along the thicknesses of the sections, wherein the section thicknesses are preferably parallel to the rotational axis. The multiple sections preferably have substantially the same mass, but can alternatively have different masses. The center of mass for each eccentric mass section is preferably offset from the mass couple connection point for each eccentric mass section, and is preferably arranged proximal an adjacent eccentric mass

section. In operation, the eccentric mass sections separate until the centers of mass of the eccentric mass sections oppose each other across the axis of rotation.

(81) When the eccentric mass **121** is cooperatively formed by multiple sections (e.g., a first eccentric/hanging mass and a second eccentric/hanging mass, etc.), the mass couple **124** (e.g., a mass coupler) preferably also includes multiple sections, wherein each mass couple section statically couples to an eccentric mass section. The mass couple sections are preferably rotatably coupled to the cam **122**, but can alternatively be statically coupled to the cam **122**. Each mass couple section is preferably rotatably coupled to the remaining mass couple sections, but can alternatively be statically coupled to one or more of the remaining mass couple sections. In one variation as shown in FIGS. **8A** and **8B**, the end of each mass couple section opposing the eccentric mass section is rotatably coupled to the housing. The angular positions of mass couple section ends are preferably static relative to the housing, wherein the mass couple section ends are preferably equally distributed about the axis of rotation. In another variation, the end of the each mass couple section opposing the eccentric mass section includes a bearing, wherein the bearing is slidably engaged within a circumferential groove statically coupled to the cam **122** and encircling the rotational axis. When the rotation frequency of the rotating surface **900** is below or above the excitation frequency for the cooperatively defined eccentric mass **121**, the centrifugal force of the rotation preferably retains the eccentric mass sections (and mass couple sections) in substantially adjacent positions. When the rotation frequency of the rotating surface **900** is at the excitation frequency, the centrifugal force preferably causes the bearings to slide within the groove, distributing the multiple eccentric mass sections substantially equally about the rotational axis. The bearings and/or the eccentric mass sections can each additionally include magnets, disposed in repulsive relation to adjacent magnets, which facilitate eccentric mass separation in response to the receipt of a system oscillation. In another variation, the mass couple sections rotatably couple along the longitudinal axis of an axle extending from the cam **122** (e.g., mass couple sections are stacked along the axle). In another variation, one mass couple section is statically connected to the cam **122** while the remaining mass couple sections are rotatably connected to the cam **121**. However, the mass couple sections can be otherwise connected to the cam **122**.

(82) When the mass couple **124** couples to the cam **122** (e.g., a positioning subcomponent) at the rotational axis, the mass couple **124** is preferably operable between the coupled mode, wherein the mass couple **124** connects the eccentric mass **121** to the cam **122**, and the decoupled mode, wherein the mass couple **124** disconnects the eccentric mass **121** from the cam **122**. In one variation, the mass couple **124** is a disk located within the lumen defined by an interior bearing surface **127** (e.g., a cylindrical inner surface) of the cam **122**, wherein the disk can rotate relative to the interior bearing surface in the decoupled mode and is coupled to the interior bearing surface by a friction element in the coupled mode. The mass couple sections are preferably rotatably coupled to the disk, but can alternatively be disk sections (e.g., concentric circles, arcuate pieces, etc.). The friction element can be a high-friction coating along the interior bearing surface, a high-friction coating along the mass couple **124** exterior, a roller or wedge, or any other suitable element capable of providing friction between the interior bearing surface and the mass couple **124**. The friction element is preferably selected such that the cooperative centrifugal force of the eccentric mass **121** in the coupled mode applies sufficient force to the mass couple **124** such that the friction between the mass couple **124** and the interior bearing surface retains the mass couple position relative to the cam **122**. The friction element is preferably selected such that the cooperative centrifugal force of the eccentric mass sections in a separated or decoupled mode does not provide enough force to interface friction to retain the mass couple position relative to the cam **122**, thereby allowing free mass couple rotation. In another variation, the mass couple **124** is rotatably mounted on an axle extending from the cam **122** by bearings, wherein the mass couple **124** can be statically coupled to the cam **122** by one or more sets of magnets or pistons extending from the adjacent broad faces of the cam **122** and mass couple **124**. However, the static mass couple connection to the cam **122** to

achieve the coupled mode can be selectively controlled by any other suitable passive or active means.

(83) The eccentric mass **121** can additionally include a connection mechanism **125** (e.g., a coupler) that functions to couple the multiple sections together. The connection mechanism is preferably located on the interfaces of adjacent sections, but can alternatively be located within the section bodies, at the interfaces of adjacent mass couple sections, or at any other suitable location. The coupling force of the connection mechanism is preferably selected such that it is substantially equal to or lower than the angular separation force experienced by the individual eccentric mass sections when the system is rotating at the excitation frequency. However, the coupling force can have any other suitable magnitude. The connection mechanism can be a mechanical connection (e.g., adhesive, clips, Velcro, etc.) with a separation force substantially equivalent to the coupling force, a magnetic connection wherein adjacent eccentric mass or mass couple sections include complimentary magnets, or any other suitable mechanism that can selectively connect adjacent eccentric mass sections together.

(84) In one alternative, the eccentric mass **121** is collectively formed by a first and a second section (e.g., the eccentric mass **121** is divided radially into two sections), wherein the first section is a reflected duplication of the second section. In operation, the first and second sections are preferably diametrically opposed and spin about the axis of rotation of the positioning mechanism when the system vibration reaches the resonance frequency of the eccentric mass **121**. In a second alternative, the eccentric mass **121** is collectively formed by a first, second, and third section with substantially the same mass, wherein the first, second and third sections are preferably substantially evenly distributed about the rotational axis when the system rotational speed reaches the resonance frequency of the eccentric mass **121**. However, the eccentric mass **121** can be formed from any number of constituent sections in any suitable configuration. Alternatively, the stabilizing mechanism **700** can be any other suitable mechanism.

(85) The FIGURES illustrate the architecture, functionality and operation of possible implementations of systems, methods and computer program products according to preferred embodiments, example configurations, and variations thereof. In this regard, each block in the flowchart or block diagrams may represent a module, segment, step, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block can occur out of the order noted in the FIGURES. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(86) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

Claims

1. A system, comprising: a pump, configured to statically mount to a wheel; a first eccentric mass rotatably mounted relative to the pump; a second eccentric mass rotatably mounted relative to the pump; and a positioning subcomponent comprising a cylindrical inner surface constraining translation of a region of the first eccentric mass element, wherein the first eccentric mass is operable between: a pumping mode, wherein the first eccentric mass is arranged proximal to the second eccentric mass and applies a pumping force to the pump, wherein the pumping force is

- generated by rotation of the pump relative to the first eccentric mass; and a nonpumping mode, wherein the first eccentric mass is separated from the second eccentric mass by an arcuate force applied to the first eccentric mass by the positioning subcomponent.
2. The system of claim 1, wherein the arcuate force is passively generated.
 3. The system of claim 1, wherein the arcuate force acts on the first eccentric mass after separation of the first eccentric mass and the second eccentric mass.
 4. The system of claim 3, wherein the positioning subcomponent is configured to maintain a predetermined angular position of the first eccentric mass.
 5. The method of claim 1, wherein the first eccentric mass is configured to rotate with the wheel in the nonpumping mode.
 6. The system of claim 1, wherein the first eccentric mass and the second eccentric mass are connected to a rotor and stator.
 7. The system of claim 6, wherein the rotor and stator are configured to dynamically maintain the first eccentric mass within a range of non-zero angles relative to the second eccentric mass in the nonpumping mode.
 8. The system of claim 6, wherein the second eccentric mass is connected to at least one of the rotor or the stator, wherein the rotor and stator are configured to harvest electrical energy during the pumping and nonpumping modes.
 9. The system of claim 1, wherein the first eccentric mass and the second eccentric mass are connected by coupler, the coupler applying an arcuate force to the first eccentric mass in the pumping mode.
 10. The system of claim 1, further comprising a processing system comprising a set of processors configured to control a system operation between the pumping mode and the nonpumping mode.
 11. The system of claim 1, wherein the arcuate force changes in magnitude based on an angular velocity of the first eccentric mass.
 12. The system of claim 1, wherein a center of mass of the first and second eccentric masses is offset from a wheel axis of rotation during the pumping mode and aligned with the wheel axis of rotation during the nonpumping mode.
 13. A system for pump actuation, comprising: a pump, configured to statically mount to a wheel; a first hanging mass and a second hanging mass, each rotatably mounted relative to the pump and operable between: a pumping mode, wherein a pumping force, generated by relative motion between the second hanging mass and the pump, is applied to the pump; and a nonpumping mode; and a coupler mounted to the first hanging mass, wherein the coupler constrains the first hanging mass relative to the second hanging mass mode in the pumping mode, and wherein the first hanging mass and the second hanging mass transition between the pumping mode and nonpumping mode responsive to a change in a rotational speed of the wheel.
 14. The system of claim 13, further comprising a generator connected to the second hanging mass, wherein the generator harvests energy from relative rotation between the wheel and at least one of: the first hanging mass or the second hanging mass.
 15. The system of claim 13, wherein the first hanging mass and the second hanging mass are connected to a stator and a rotor.
 16. The system of claim 13, wherein a centrifugal force separates the first hanging mass from the second hanging mass to transition the system from the pumping mode to the nonpumping mode.
 17. The system of claim 13, wherein the system is configured to transition the first hanging mass and the second hanging mass from the pumping mode to the nonpumping mode when the wheel rotation reaches a threshold velocity.
 18. The system of claim 13, further comprising a force mechanism coupled to the first hanging mass and configured to control motion of the first hanging mass in the nonpumping mode.
 19. The system of claim 18, wherein the force mechanism controls motion of the first hanging mass

by applying an arcuate force.

20. The system of claim 18, wherein the force mechanism comprises a spring.
